

ERBENT, Turkmenistan

WMO: 386560

Lat: 39.317N

Lon: 58.600E

Elev: 285

StdP: 14.54

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.1	18.1	-1.5	5.1	16.8	5.6	7.4	23.9	17.1	42.2	14.1	43.7	3.7	90	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.6	108.2	70.2	105.7	69.7	103.3	68.9	76.3	99.7	73.9	97.6	71.7	97.1	5.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.1	104.3	93.7	65.3	94.3	90.9	62.0	84.1	87.7	40.0	100.1	37.7	97.0	35.6	96.8	94.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	13.1	11.2	6.3	112.2	10.0	2.4	-0.9	113.9	-6.8	115.3	-12.4	116.6	-19.7	118.3
			5.3	78.4	9.6	5.4	-1.6	82.2	-7.3	85.4	-12.6	88.4	-19.6	92.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	63.6	36.6	40.2	52.6	64.6	77.3	86.5	90.7	87.5	76.2	62.4	48.2	38.5	(6)	
(7)		DBStd	20.81	9.19	10.18	8.93	9.09	6.59	5.56	4.94	5.51	6.81	8.51	9.28	8.15	(7)	
(8)		HDD50	1305	419	291	74	7	0	0	0	0	11	136	367		(8)	
(9)		HDD65	3570	879	694	394	115	4	0	0	5	148	508	823		(9)	
(10)		CDD50	6267	4	17	156	446	847	1096	1263	1164	786	397	82	9	(10)	
(11)		CDD65	3058	0	0	11	104	386	646	798	699	341	69	4	0	(11)	
(12)		CDH74	45093	0	1	165	1160	4789	9968	13196	10839	4164	770	40	1	(12)	
(13)		CDH80	27756	0	0	49	470	2482	6327	8986	7081	2098	254	9	0	(13)	
(14)	Wind	WSAvg	4.6	4.1	4.8	5.3	5.3	4.8	5.0	4.6	4.1	4.3	4.1	3.9	4.3	(14)	
(15)	Precipitation	PrecAvg	4.2	0.5	0.4	0.7	0.8	0.5	0.1	0.1	0.0	0.1	0.2	0.4	0.5	(15)	
(16)		PrecMax	8.0	1.4	1.5	2.9	2.9	2.4	0.4	0.7	0.2	0.6	1.1	1.2	1.9	(16)	
(17)		PrecMin	2.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
(18)		PrecStd	1.3	0.4	0.4	0.6	0.7	0.6	0.1	0.2	0.0	0.1	0.2	0.3	0.4	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.3	70.4	86.4	96.7	102.6	108.9	112.5	110.7	103.2	92.6	79.5	66.7	(19)	
(20)		MCWB	48.2	54.3	58.3	64.7	66.9	72.0	71.6	70.2	67.7	63.7	59.4	50.7		(20)	
(21)		2%	DB	57.2	63.7	78.2	90.1	98.2	105.6	109.2	107.1	98.3	86.4	71.0	58.9	(21)	
(22)		MCWB	45.9	49.7	56.2	62.0	66.7	69.9	70.1	69.7	66.1	60.8	54.4	47.7		(22)	
(23)		5%	DB	52.1	58.6	72.5	85.4	95.1	103.2	106.5	104.4	94.6	82.2	66.2	53.7	(23)	
(24)		MCWB	43.6	47.0	54.1	60.4	65.2	68.3	69.9	68.8	65.4	59.2	52.2	45.0		(24)	
(25)		10%	DB	48.3	54.0	67.4	80.8	91.8	100.5	103.9	101.7	90.9	77.9	62.1	49.4	(25)	
(26)		MCWB	41.5	45.0	52.2	58.9	64.2	67.5	69.8	67.9	63.6	57.0	50.4	42.5		(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.7	54.8	59.9	66.8	72.6	77.7	79.9	78.4	75.5	66.9	60.2	52.5	(27)	
(28)		MCDWB	59.2	67.2	81.0	88.8	93.3	102.7	103.1	99.1	96.2	84.7	78.1	64.3		(28)	
(29)		2%	WB	47.1	50.8	57.4	64.3	69.3	74.4	77.1	74.4	70.0	62.9	56.0	48.5	(29)	
(30)		MCDWB	55.5	61.3	74.8	85.5	91.7	98.5	101.0	97.2	91.0	82.2	67.9	57.7		(30)	
(31)		5%	WB	44.3	47.9	55.4	62.0	67.3	72.3	74.6	71.8	67.0	60.4	53.4	45.5	(31)	
(32)		MCDWB	51.9	57.0	70.3	82.0	89.8	96.7	98.8	98.2	88.9	78.6	64.8	52.4		(32)	
(33)		10%	WB	41.7	45.4	53.0	59.8	65.6	70.0	72.1	69.5	64.9	58.1	50.8	43.3	(33)	
(34)		MCDWB	47.8	53.6	65.8	77.7	87.8	95.1	97.8	98.8	87.8	75.5	61.1	48.7		(34)	
(35)	Mean Daily Temperature Range	MDBR	15.7	17.3	19.9	20.6	22.6	23.1	22.6	24.3	24.3	23.3	18.8	14.9		(35)	
(36)		5% DB	MCDBR	22.8	25.3	28.2	27.4	26.7	26.4	25.6	27.2	28.1	28.1	24.8	22.3		(36)
(37)		MCWBR	13.5	14.1	12.2	11.2	11.0	10.7	10.1	11.5	12.8	13.8	12.8	13.8		(37)	
(38)		5% WB	MCDBR	21.4	23.5	25.5	24.6	23.4	23.0	21.6	23.9	24.4	25.3	22.7	20.2		(38)
(39)		MCWBR	13.0	13.8	11.7	11.1	11.1	12.1	12.1	13.1	13.4	13.7	12.7	12.8		(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.408	0.453	0.516	0.566	0.555	0.563	0.564	0.516	0.487	0.486	0.440	0.407		(40)	
(41)		taud	1.998	1.881	1.775	1.719	1.777	1.761	1.770	1.854	1.891	1.871	1.958	1.997		(41)	
(42)		Ebn at Noon	225	232	232	231	237	235	233	241	238	219	211	213		(42)	
(43)		Edh at Noon	40	52	64	72	69	71	69	62	56	52	42	38		(43)	
(44)	All-Sky Solar Radiation	RadAvg	673	953	1357	1781	2175	2502	2423	2238	1828	1261	830	600		(44)	
(45)		RadStd	65	102	83	143	102	69	77	81	54	78	59	78		(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	+1.15	N/A	N/A	N/A	N/A	N/A	+148	(47)

Nomenclature: See separate page